Session Proposal

High Level explanation:

A session is intended to show a biking session to the user to compartmentalize incidents in an easy-to-understand way. A session represents a period where Incidents can occur. If there is an active connection with the IBDC device then a session is active. It will start with device connection and end with the last device communication for 2+ hours. This has the effect of making any gap in connectivity of 2 hours or more drive a separation of sessions. Any gap of less than 2 hours is believed to be part of the same session. There will be no minimum session length as a rider may travel down a roadway for mere minutes testing out how a new component of their bike feels or checking the tires.

Session Display

We will show “No Incidents” on any session with no incidents. They will still be tracked even if no incidents exist to allow the user to better recognize their past riding records. We will not allow the deletion of any session with incidents associated with it. Once all incidents are deleted from a session then a session may be deleted by the user. Any session without incident data needs to present an option for the user to delete it. This allows the users to clean up erroneous data.

We won’t track or display the session end time or duration as it won’t be accurate in most cases and presenting inaccurate or irrelevant data detracts from the credibility of the application.

Any incident can be deleted at will. There should be a confirmation window and then data should NOT be retained as this adds added complexity and doesn’t match the expected behavior for what the user selected. (I.E. The user clicked delete data but its retained - With potentially large numbers of images this is a valid space concern for some users)

Technical Proposed Process:

There should be a persistent “Active Session” in memory when the application is running. This will show along with historical sessions on the session list screen. This active session will be stored separately in its own relation/table in the database.

Type Parameters:

Incidents[]
LastGPSCoordinate
LastGPSTime
StartDateStamp

When communication with the IBDC device occurs, it indicates a device connection,

If ActiveSession.LastGPSTime is null or the difference between current time and LastGPSTime is >= 2 hours, a new session will start. That means:

- If LastGPSTime != null then a previous session existed. This means data from the active session should be added as a new session object to the list of historical sessions in the database. ActiveSession starts representing a new session so activeSession LastGPSCoordinate, LastGPSTime, and StartDateStamp all update based on the current GPS location and the incident array is cleared.
- If LastGPSTime == null then activeSession LastGPSCoordinate, LastGPSTime, and StartDateStamp all update based on the current GPS location and the incident list is validated to be empty. This indicates a brand-new session for the application.

If ActiveSession.LastGPSTime is NOT null and the LastGPSTime was less than 2 hours ago then we are mid-session. LastGPSTime and LastGPSCoordinate will be updated in the ActiveSession object.

The reason for storing every known device connection time (Last GPSTime) is to track the duration of any disconnection. The reason for StartDateStamp is to track the day the session occurred on and order multiple sessions for a day properly. The reason for LastGPSCoordinate is to have something for the user in the event an incident is triggered during a disconnection. Based on the time we can decide if the information is valuable or not.

Please ask if you have any questions or concerns.

After team members review this document, I would like to email this to our sponsor for input/validation. This will let me get buy-in from all stakeholders before coding the logic.